Systems Analysis and Control

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Lecture 2: Feedback, Differential Equations, and State-Space

Introduction

In this Lecture, you will learn:

1. A Classic Illustration of Feedback
2. Modeling using differential equations.
3. The State-Space Framework.

PART 1: An Example of Control without Dynamics

Just for Motivation: Makes it easier to solve for the output

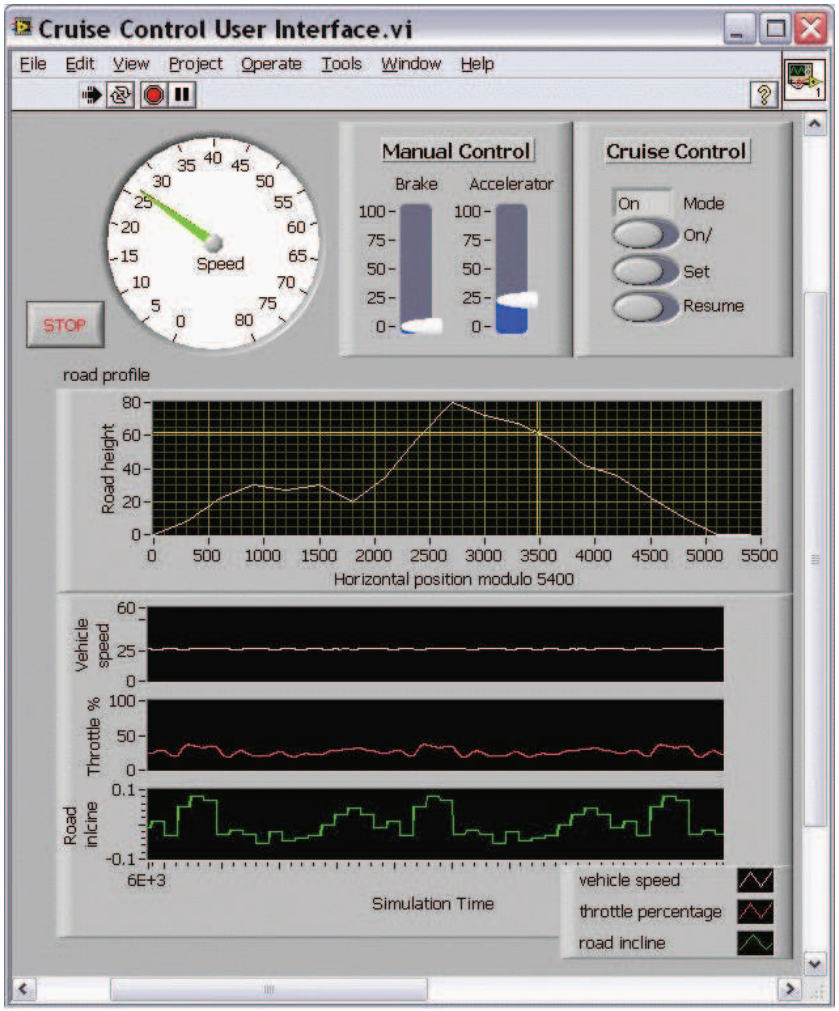
CRUISE CONTROL

Plant: The Automobile (Car)

- **Input:** Throttle Position, $\theta_{throttle}$.
- **Output:** Real Velocity, $\mathbf{v_{true}}$.
- **Plant:** No Dynamics! Speed is proportional to throttle (proportional gain).

$$\underbrace{\mathbf{v_{true}}}_{\text{output}} = \underbrace{10}_{\text{Plant}} \cdot \underbrace{\theta_{throttle}}_{\text{input}}$$

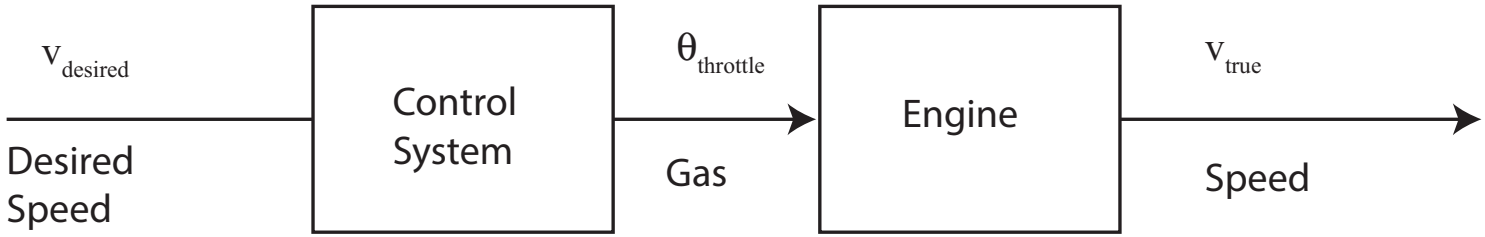
The *gain factor* is $10\ mph/^{\circ}$.



Cruise Control: Open-Loop vs. Closed-Loop

Open Loop Control

First lets start with open loop control



Actuator: Throttle

Controller:

- **Input:** Desired Velocity, $\mathbf{v_{desired}}$.
- **Output:** Throttle, $\theta_{throttle}$.

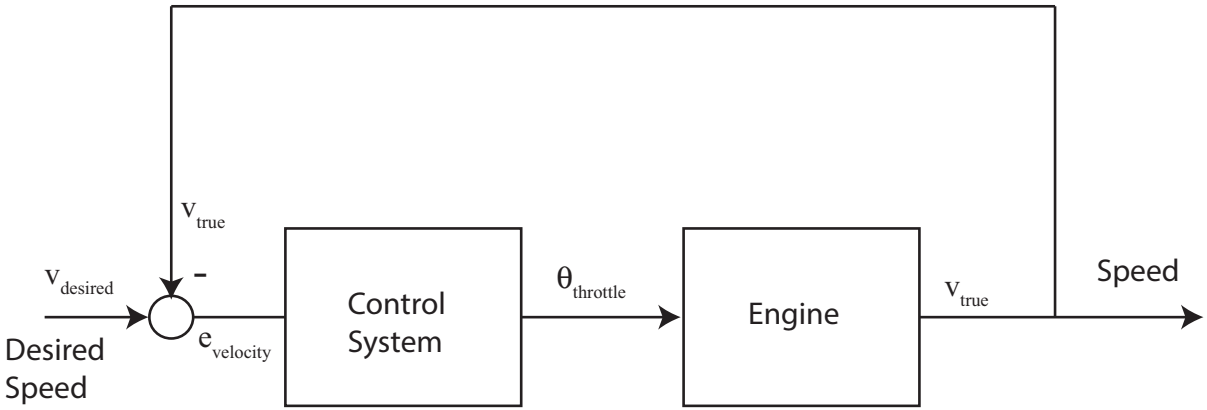
Open Loop Controller We use a simple controller based on our knowledge of the plant.

$$\underbrace{\theta_{throttle}}_{\text{output}} = \underbrace{\frac{1}{10}}_{\text{controller}} \underbrace{\mathbf{v_{desired}}}_{\text{input}}$$

Cruise Control: Open-Loop vs. Closed-Loop

Closed Loop Control: The Error signal

Now lets try using closed loop control



Actuator: Throttle

Sensor: Speedometer

Look at the **CONTROLLER:**

- **Input:** Error in Velocity, $e_{velocity} = \mathbf{v_{true}} - \mathbf{v_{desired}}$.
- **Output:** Throttle, $\theta_{throttle}$.

Proportional Feedback: Amplify the error signal by a scalar *gain* k.

$$\underbrace{\theta_{throttle}}_{\text{output}} = -k \cdot e_{velocity} = \underbrace{-k}_{\text{control}} \cdot \underbrace{\left(\mathbf{v_{true}} - \mathbf{v_{desired}}\right)}_{\text{input}}$$

Closed Loop vs. Open Loop (Solving for $\mathbf{v_{true}}$)

Open Loop: Two equations:

$$\mathbf{v_{true}} = 10 \cdot \theta_{throttle} \qquad \text{and} \qquad \theta_{throttle} = \frac{1}{10}\mathbf{v_{desired}}$$

Combining, we get

$$\mathbf{v_{true}} = 10\frac{1}{10}\mathbf{v_{desired}} = \mathbf{v_{desired}}.$$

Open-loop control has **No Error!**

Closed Loop: Two equations:

$$\mathbf{v_{true}} = 10 \cdot \theta_{throttle} \qquad \text{and} \qquad \theta_{throttle} = -k\left(\mathbf{v_{true}} - \mathbf{v_{desired}}\right).$$

Combining these, we get $\mathbf{v_{true}} = -10 \cdot k(\mathbf{v_{true}} - \mathbf{v_{desired}})$.
Lets **Choose** $k = 10$.

$$\mathbf{v_{true}} = \frac{10 \cdot k}{1 + 10 \cdot k}\mathbf{v_{desired}} = \frac{100}{101}\mathbf{v_{desired}} = .99\mathbf{v_{desired}}.$$

Closed-loop control has **1% Error**.

Impact of Error and Disturbances

Comparison:

- Open Loop: No error
- Closed Loop: Small error
  - Error gets very small if $k \to \infty$, since

$$\mathbf{v_{true}} = \frac{10 \cdot k}{1 + 10 \cdot k} \mathbf{v_{desired}} \to \mathbf{v_{desired}}.$$

Question: Why use feedback?

- **Answer:** Life is Messy.

Problems:

- Modeling Errors: What if our model is wrong by 10%, so

$$\mathbf{v_{true}} = 11 \cdot \theta_{throttle}$$

- Disturbances: An Incline of $i_{disturbance}$ degrees will reduce the throttle by $.5/^\circ$.

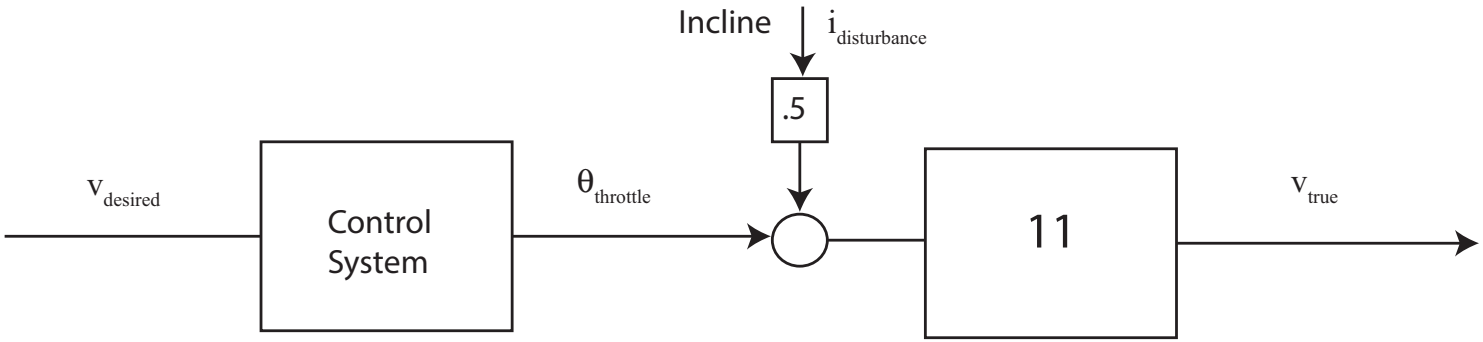
$$\Delta\theta_{throttle} = -.5 \cdot i_{disturbance}$$

Impact of Error and Disturbances

Open Loop

Let $\mathbf{v_{desired}} = 50mph$, $i_{disturbance} = -1°$.

Now Recalculate the *Open Loop* Output : $\mathbf{V_{true}}$



$$\mathbf{v_{true}} = 11(\theta_{throttle} - .5 \cdot i_{disturbance})$$

$$\theta_{throttle} = \frac{1}{10}\mathbf{v_{desired}} = 5$$

we have

$$\mathbf{v_{true}} = 11(5 + .5) = 60.5mph$$

Which is **NOT ACCEPTABLE!!!**.

Impact of Error and Disturbances

Closed Loop

Recalculate the *Closed Loop* output : $\mathbf{v_{true}}$

- **Plant with Disturbance:** $\mathbf{v_{true}} = 11 \cdot (\theta_{throttle} - .5 \cdot i_{disturbance})$
- **Controller:** $\theta_{throttle} = -k\left(\mathbf{v_{true}} - \mathbf{v_{desired}}\right) = -k(\mathbf{v_{true}} - 50)$

Combine these equations and solve for $\mathbf{v_{true}}$.

$$\mathbf{v_{true}} = 11(-k \cdot \mathbf{v_{true}} + 50 \cdot k + .5) = -11 \cdot k \cdot \mathbf{v_{true}} + 11 \cdot 50 \cdot k + 5.5$$

Solving for $\mathbf{v_{true}}$ yields

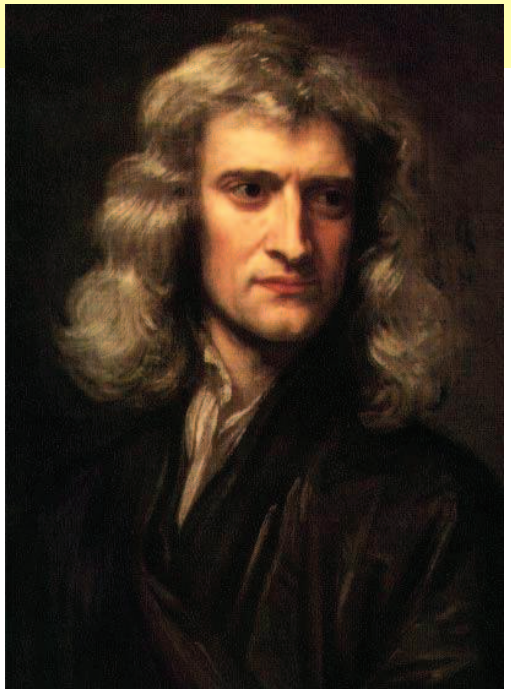
$$\mathbf{v_{true}} = \frac{11k + .11}{1 + 11k} 50 = \frac{110.11}{111} 50 = .991 \cdot 50 = 49.6mph$$

Note: If you make k REALLY big, then the error gets REALLY small.

- In dynamic models, big k can *destabilize* the closed-loop system.

PART 2: A Brief Review of Modeling

Ordinary Differential Equations (ODEs)



Models can be

- static ($x = Ku$).
- dynamic ($\dot{x} = -x + u$).

Physics-based Modeling

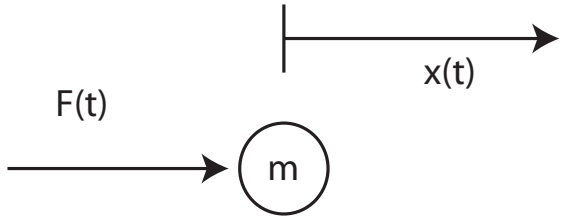
- Mechanics and circuits define ODEs
- Identify *states* (position, voltage, et c.)
- Identify governing differential equations.

Newton invented ODE models in 1684:

- Newton's Second Law: (x is position)

$$\frac{d^2}{dt^2}x(t) = F(t)/m$$

- $x(t)$ is a state (can also be a signal (output))
- $F(t)$ is a signal (input)

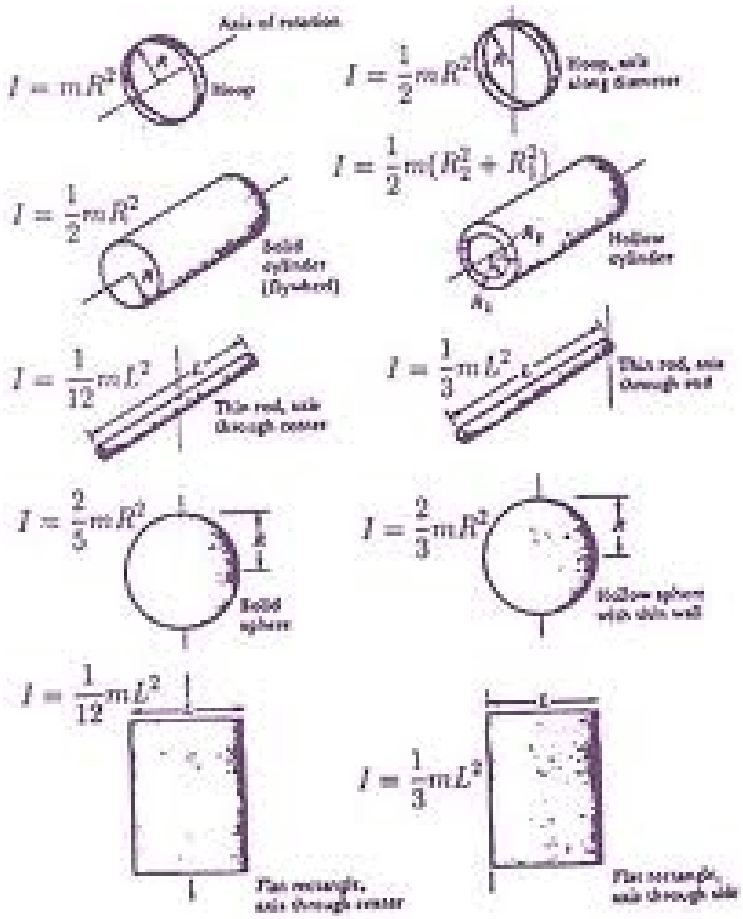


Newton's Second Law Applied to Rigid Bodies

The rate of change of angular momentum is given by

$$\sum M(t) = I\alpha(t) = I\ddot{\theta}(t)$$

- $\alpha(t) = \ddot{\theta}(t)$ is the angular acceleration.
- $\theta(t)$ is the state (angular displacement)
- I is the moment of inertia.
- $M(t)$ is the torque (moment).

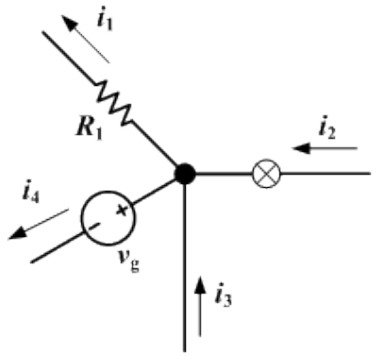


Other Sources of Differential Equations

Voltage Laws

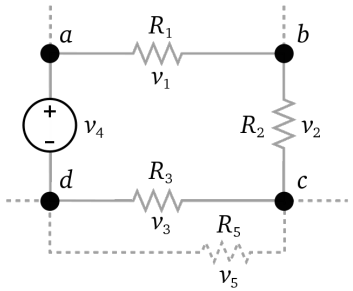
Kirchhoff's Current Law (KCL):
Current is conserved at each junction

$$\sum_k i_k(t) = 0$$



Kirchhoff's Voltage Law (KVL): Net
Voltage change around any loop is zero.

$$\sum_k V_k(t) = 0$$

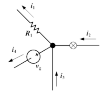


These are combined with standard voltage laws such as voltage drop across a resister, inductor and capacitor:

$$V_r(t) = Ri_r(t) \qquad \frac{d}{dt}i_L(t) = \frac{1}{L}V_L(t) \qquad \frac{d}{dt}V_c(t) = \frac{1}{C}i_c(t)$$

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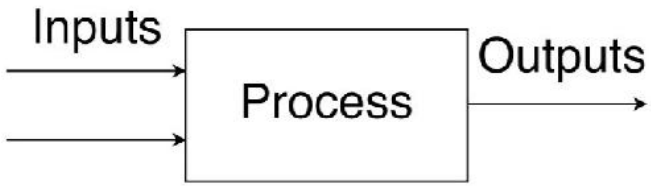
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Component	Voltage–current	Current–voltage	Voltage–charge
 Capacitor	$v(t) = \frac{1}{C}\int_0^1 i(\tau)d\tau$	$i(t) = C\frac{dv(t)}{dt}$	$v(t) = \frac{1}{C}q(t)$
 Resistor	$v(t) = Ri(t)$	$i(t) = \frac{1}{R}v(t)$	$v(t) = R\frac{dq(t)}{dt}$
 Inductor	$v(t) = L\frac{di(t)}{dt}$	$i(t) = \frac{1}{L}\int_0^1 v(\tau)d\tau$	$v(t) = L\frac{d^2q(t)}{dt^2}$

Review: Modeling

Differential Equations



Coupled Diff Eqs:

$$\dot{x}(t) = ax(t) + bz + eu(t)$$
$$\dot{z}(t) = cx(t) + dz(t) + fu(t)$$
$$y(t) = gx(t) + hz(t) + iu(t)$$

Higher Order Diff Eqs:

$$\dddot{x}(t) = a\ddot{x}(t) + b\dot{x}(t) + cx(t) + eu(t)$$
$$y(t) = f\ddot{x}(t) + g\dot{x}(t) + hx(t) + iu(t)$$

Where

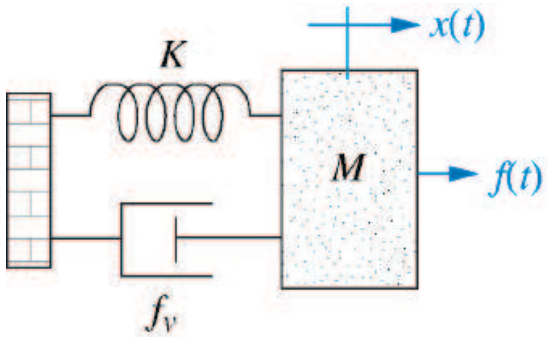
- $u(t)$ is the Input
- $y(t)$ is the Output
- $x(t), z(t)$ are the *state variables*.
  - position, heading, velocity, etc.

Dynamic Model: Suspension System

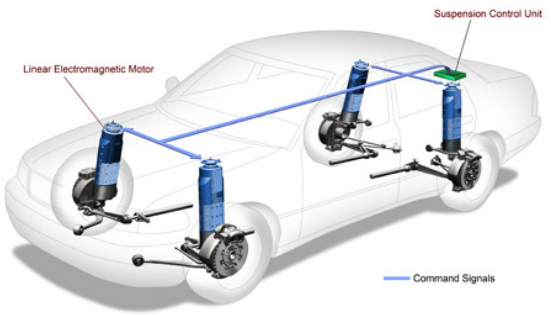
Mass-Spring Model

We wish to study the motion of the vehicle subject to disturbances.

- Model the car as a solid mass
- Control the vertical motion of the car ($x(t)$)



(a)



Inputs: Force, $f(t)$.
Outputs: Displacement, $y(t) = x(t)$.

Definition 1.

A system with one input and one output is Single-Input, Single-Output (**SISO**).
A system with more than one input *or* more than one output is Multi-Input Multi-Output (**MIMO**)

Dynamic Model: Suspension System

Mass-Spring Model

Plant Dynamics: What are the Forces on the mass?

- Input Force: $f(t)$
- Spring Force: Opposes motion in x with spring constant K.

$$F_s(t) = -Kx(t)$$

- Damper Force: Opposes motion in $\dot{x}$ with damping coefficient c

$$F_d(t) = -c\dot{x}(t)$$

- Now apply Newton's Second Law:

$$m\ddot{x}(t) = F_s(t) + F_d(t) + f(t)$$

System Model:

$$\ddot{x}(t) = -\frac{K}{m}x(t) - \frac{c}{m}\dot{x}(t) + \frac{1}{m}f(t)$$
$$y(t) = x(t)$$

Standard Forms

State-Space and Transfer Functions

Once we have our dynamic model

$$\ddot{x}(t) = -\frac{K}{m}x(t) - \frac{c}{m}\dot{x}(t) + \frac{1}{m}f(t) \qquad \text{Differential Equations}$$

$$y(t) = x(t) \qquad \text{Output Equation}$$

This model can be expressed in two standard forms

- Transfer Function
- State-Space

We will discuss these in more depth soon. For now:

Transfer Function: Apply the Laplace Transform to both equations and solve for the output.

$$s^2\hat{x}(s) = -\frac{K}{m}\hat{x}(s) - \frac{c}{m}s\hat{x}(s) + \frac{1}{m}\hat{f}(s) \qquad \text{Differential Equations}$$

$$\hat{y}(s) = \hat{x}(s) \qquad \text{Output Equation}$$

which yields

$$\hat{y}(s) = \frac{1}{ms^2 + cs + K}\hat{f}(s)$$

PART 3: State-Space Formulation

A Standard Form for writing Diff. Eqns.

State-Space is a way of writing first order differential equation using matrices.

$$\dot{\vec{x}}(t) = A\vec{x}(t)$$

where $\vec{x}$(t) is a vector and $A \in \mathbb{R}^{n \times n}$ is a square matrix.

Example:

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Is equivalent to writing the three differential equations

$$\begin{aligned} \dot{x}_1 &= -x_1 + x_3 \\ \dot{x}_2 &= 2x_1 \\ \dot{x}_3 &= -x_2 + x_3 \end{aligned}$$

Writing equations in state-space has many advantages

Review: Equations of Motion

Multiple Variables and State-Space

Consider the system

$$\dot{x} = ax + by$$
$$\dot{y} = cx + dy$$

When we have multiple coupled equations: **Convert to State-Space:**

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

Which is easily expressed as

$$\dot{\mathbf{x}} = A\mathbf{x}$$

where

- $\mathbf{x}$ is a vector.
- A is a matrix.

State-Space Form for Systems

Definition 2.

State-Space Form is a convenient way of representing *linear* multivariate or MIMO systems using 4 matrices.

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

- u is the vector of **Inputs**.
- y is the vector of **Outputs**.
- x is the **State**.

$u \in \mathbb{R}^m$, $y \in \mathbb{R}^p$, and $x \in \mathbb{R}^n$ can be vectors of any dimension. However, the matrices must be compatable (the right size):

$$A \in \mathbb{R}^{n \times n} \qquad B \in \mathbb{R}^{n \times m}$$
$$C \in \mathbb{R}^{p \times n} \qquad D \in \mathbb{R}^{p \times m}$$

- $u \in \mathbb{R}^m$ means u is a real vector of length m.
- $C \in \mathbb{R}^{p \times n}$ means C is a matrix with p rows and n columns.

Putting Things in State-Space Form

Reducing Higher Order Dynamics

When we have higher order derivatives,

$$\dddot{x}(t) = ax(t) + b\dot{x}(t) + c\ddot{x}(t) + u(t)$$
$$y(t) = x(t) + u(t)$$

we can put it in state-space form by

- Introducing new variables.

Procedure:

- Define a new variable for every derivative term except for the the highest order one.
  - e.g. Let $x_1 = x$, $x_2 = \dot{x}$ and $x_3 = \ddot{x}$.
- Add a new first order differential equation for each new variable.
  - e.g. $\dot{x}_1 = x_2$ and $\dot{x}_2 = x_3$
- Then put in state-space form.

So, for example, we would have 3 equations

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = x_3(t)$$
$$\dot{x}_3(t) = ax_1(t) + bx_2(t) + cx_3(t) + u(t)$$

Putting Things in State-Space Form

Using our first-order equations:

$$\dot{x}_1(t) = x_2(t); \qquad \dot{x}_2(t) = x_3(t)$$
$$\dot{x}_3(t) = ax_1(t) + bx_2(t) + cx_3(t) + u(t) \qquad y(t) = x_1(t) + u(t)$$

We construct the state-space representation:

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

as

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \overbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a & b & c \end{bmatrix}}^{A} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \overbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}^{B} u(t)$$

$$y(t) = \underbrace{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}}_{C} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \end{bmatrix}}_{D} u(t)$$

Putting Things in State-Space Form
Using our first-order equations:
$\dot{x}_1(t) = x_2(t);$ $\dot{x}_2(t) = x_3(t)$
$\dot{x}_3(t) = ax_1(t) + bx_2(t) + cx_3(t) + u(t)$ $y(t) = x_1(t) + u(t)$
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$y(t) = \underbrace{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}}_{C} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \underbrace{[1]}_{D} u(t)$

Spring-Mass Model:

$$\ddot{x}(t) = -\frac{K}{m}x(t) - \frac{c}{m}\dot{x}(t) + \frac{1}{m}f(t)$$
$$y(t) = x(t)$$

first-order equations:

$$\dot{x}_1(t) = x_2(t); \qquad \dot{x}_2(t) = -\frac{K}{m}x_1(t) - \frac{c}{m}x_2(t) + \frac{1}{m}f(t) \qquad y(t) = x_1(t)$$

State-Space

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{K}{m} & -\frac{c}{m} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} f(t)$$
$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Suspension System with Wheel Dynamics

Modelling a 2-mass system using Free-Body Diagrams

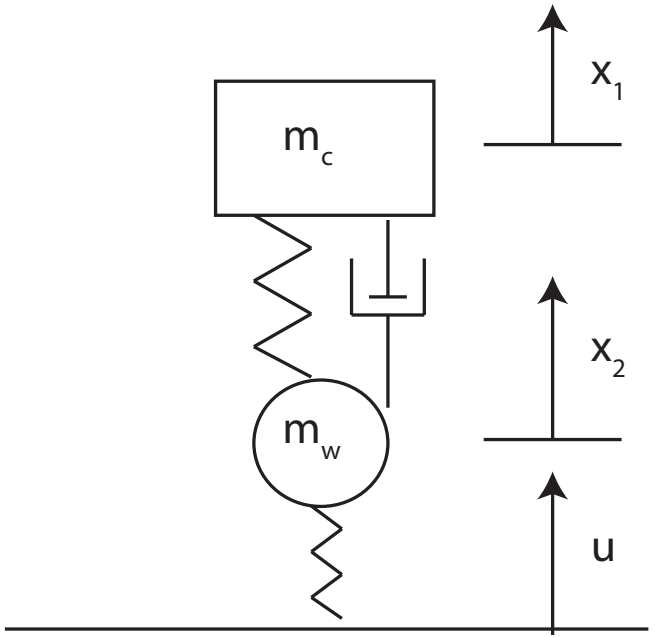
Mass-Spring systems are too easy
Lets add the dynamics of the wheel.

There are two states:
States:

- Vehicle Position, x_1
- Wheel Position, x_2

Our Input is the position of the surface of the road.
Inputs:

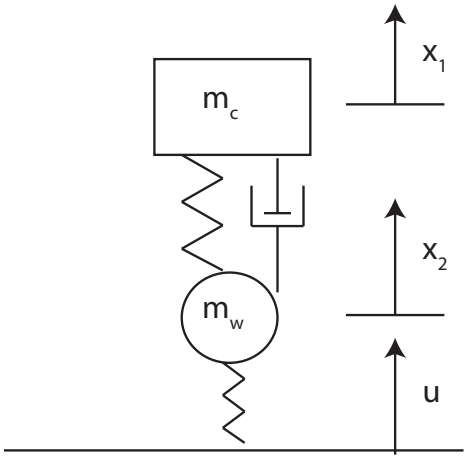
- Road Surface, u



XXXX

Suspension Model: Free Body 1

This time we write the dynamics of both the wheel and the car.

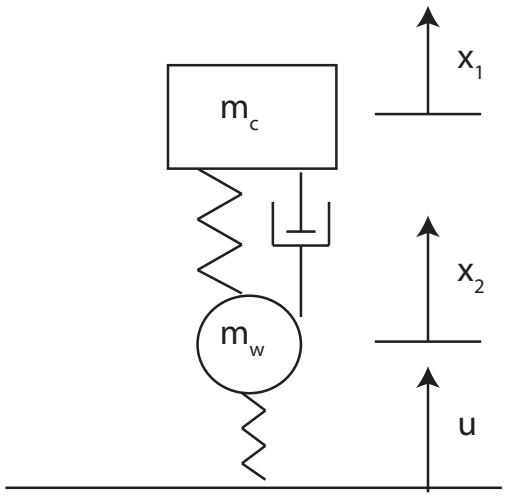


Car Dynamics: Equations of Motion

- Spring 1 Force on Car: $F_{s1,c}(t) = -K_1(x_1(t) - x_2(t))$
- Damper Force on Car: $F_{d,c}(t) = -c(\dot{x}_1(t) - \dot{x}_2(t))$
- Newton's Second Law:

$$\begin{aligned} m_c\ddot{x}_1(t) &= F_{s1,c}(t) + F_{d,c}(t) \\ &= -K_1(x_1(t) - x_2(t)) - c(\dot{x}_1(t) - \dot{x}_2(t)) \end{aligned}$$

Suspension Model: Free Body 2



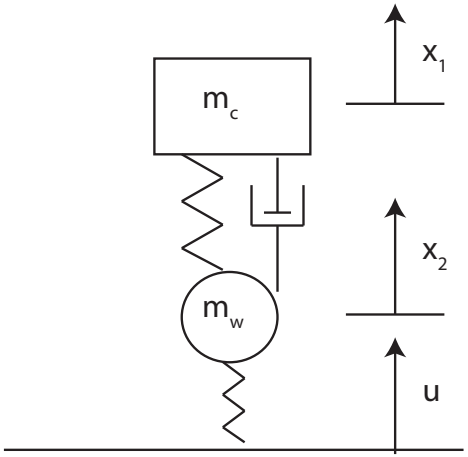
Wheel Dynamics: Equations of Motion

- Spring 1 Force on Wheel: $F_{s1,w}(t) = K_1(x_1(t) - x_2(t))$
- Spring 2 Force on Wheel: $F_{s2,w}(t) = -K_2(x_2(t) - u(t))$
- Damper Force on Wheel: $F_{d,w}(t) = c(\dot{x}_1(t) - \dot{x}_2(t))$
- Newton's Second Law:

$$\begin{aligned} m_w\ddot{x}_2(t) &= F_{s1,w}(t) + F_{s2,w}(t) + F_{d,w}(t) \\ &= K_1(x_1(t) - x_2(t)) - K_2(x_2(t) - u(t)) + c(\dot{x}_1(t) - \dot{x}_2(t)) \end{aligned}$$

Equations of Motion

Combining the dynamics, we get the coupled system dynamics.



$$m_w \ddot{x}_2(t) = K_1(x_1(t) - x_2(t)) - K_2(x_2(t) - u(t)) + c(\dot{x}_1(t) - \dot{x}_2(t))$$
$$m_c \ddot{x}_1(t) = -K_1(x_1(t) - x_2(t)) - c(\dot{x}_1(t) - \dot{x}_2(t))$$
$$y(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

This is quite complicated.

- To simplify, we would like to use a *Standard Form*.

State-Space Form: Suspension System

Recall the dynamics:

$$m_w \ddot{x}_2(t) = K_1(x_1(t) - x_2(t)) - K_2(x_2(t) - u(t)) + c(\dot{x}_1(t) - \dot{x}_2(t))$$
$$m_c \ddot{x}_1(t) = -K_1(x_1(t) - x_2(t)) - c(\dot{x}_1(t) - \dot{x}_2(t))$$
$$y(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Define the new variables z_i

$$z_1(t) = x_1(t) \qquad z_2(t) = \dot{x}_1(t) \qquad z_3(t) = x_2(t) \qquad z_4(t) = \dot{x}_2(t)$$

Which yields the following set of equations: $y(t) = \begin{bmatrix} z_1(t) \\ z_3(t) \end{bmatrix}$,

$$\dot{z}_1(t) = z_2(t)$$
$$\dot{z}_2(t) = -\frac{K_1}{m_c}(z_1(t) - z_3(t)) - \frac{c}{m_c}(z_2(t) - z_4(t))$$
$$\dot{z}_3(t) = z_4(t)$$
$$\dot{z}_4(t) = \frac{K_1}{m_w}(z_1(t) - z_3(t)) - \frac{K_2}{m_w}(z_3(t) - u(t))) + \frac{c}{m_w}(z_2(t) - z_4(t))$$

Constructing State-Space Systems

$$\dot{z}_1(t) = z_2(t)$$

$$\dot{z}_2(t) = -\frac{K_1}{m_c} z_1(t) - \frac{c}{m_c} z_2(t) + \frac{K_1}{m_c} z_3(t) + \frac{c}{m_c} z_4(t)$$

$$\dot{z}_3(t) = z_4(t)$$

$$\dot{z}_4(t) = \frac{K_1}{m_w} z_1(t) + \frac{c}{m_w} z_2(t) - \left(\frac{K_1}{m_w} + \frac{K_2}{m_w}\right) z_3(t) - \frac{c}{m_w} z_4(t) + \frac{K_2}{m_w} u(t)$$

$$y(t) = \begin{bmatrix} z_1(t) \\ z_3(t) \end{bmatrix}$$

$$\begin{bmatrix} \dot{z}_1(t) \\ \dot{z}_2(t) \\ \dot{z}_3(t) \\ \dot{z}_4(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K_1}{m_c} & -\frac{c}{m_c} & \frac{K_1}{m_c} & \frac{c}{m_c} \\ 0 & 0 & 0 & 1 \\ \frac{K_1}{m_w} & \frac{c}{m_w} & -\left(\frac{K_1}{m_w} + \frac{K_2}{m_w}\right) & -\frac{c}{m_w} \end{bmatrix} \begin{bmatrix} z_1(t) \\ z_2(t) \\ z_3(t) \\ z_4(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{K_2}{m_w} \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} z_1(t) \\ z_2(t) \\ z_3(t) \\ z_4(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u(t)$$

State-Space Representation in Matlab

Once you have a state-space representation, studying it in MATLAB is easy.
Example:

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} f(t) \qquad y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

1. Enter the matrices A, B, C, D.
   1.1 `A=[0 1; -1 -1]; B=[0;1]; C=[1 0]; D=0`
2. Declare the system object `sys` using `ss`.
   2.1 `sys=ss(A,B,C,D)`
3. Simulate the response to an impulse
   3.1 `impulse(sys)`
4. Simulate the response to a step input
   4.1 `step(sys)`

Summary

What have we learned today?

A Static Model of Cruise-Control

- Simple static model and Control
- Open Loop Control
- Closed Loop Control
- Benefits of Feedback

Dynamic Models

- Including Inputs and Outputs
- Using Newton's Laws
- MIMO and SISO systems
- Other sources of models (Kirchhoff's Laws)

State-Space

- State-Space Form